

Numbers

15 Interview Questions — Easy to Hard

The Complete DSA + SQL Interview Preparation Series

Book 3 of 13

For placement preparation, coding interviews, and product-based company rounds.

About This Book

This book covers number theory and mathematical logic questions, a category interviewers love because it tests whether you understand what's happening under the hood rather than just calling a library function. Expect primes, digit manipulation, modular arithmetic, and classic sequences.

As with Book 2, each question here is kept short: just the problem and a sample input/output pair. No constraints tables, no hints, no complexity notes — read it the way you'd hear it in a live interview and start reasoning.

How To Use This Book

Read each problem once. Restate it in your own words before writing any code. Use the sample input/output only to confirm your understanding of the expected behavior — not as a spec to reverse-engineer from. Try to solve every problem in your head or on paper before typing anything.

Difficulty Breakdown

Difficulty	Count
Easy	2
Medium	8
Hard	5

Table of Contents

Q1.	Is This Server Port Prime?	Easy
Q2.	The Self-Proud Number	Easy
Q3.	The Balanced Divisor Sum	Medium
Q4.	Does This ID Loop to Happiness?	Medium
Q5.	Flip the Transaction Code	Easy
Q6.	Split the Bill Evenly	Medium
Q7.	Sync the Delivery Cycles	Medium
Q8.	Flag Every Prime Under the Limit	Medium
Q9.	Compute Interest, Fast	Medium
Q10.	Build the Seating Triangle	Easy
Q11.	The Efficient Growth Sequence	Medium
Q12.	Translate the Invoice Number	Medium
Q13.	Decode the Antique Ledger	Medium
Q14.	The Unbreakable Encryption Check	Hard
Q15.	Count the Trailing Vault Zeros	Hard

Q1. Is This Server Port Prime?

Easy

Topic: Prime Numbers

A network config tool flags port numbers that are prime, since prime ports are reserved for internal diagnostics. Write a check that tells whether a given number is prime.

INPUT

29

OUTPUT

true

Q2. The Self-Proud Number

Easy

Topic: Armstrong Numbers

A number-puzzle app looks for 'self-proud' numbers — numbers equal to the sum of their own digits each raised to the power of the digit count. Check whether a given number qualifies.

INPUT

153

OUTPUT

true ($1^3 + 5^3 + 3^3 = 153$)

Q3. The Balanced Divisor Sum

Medium

Topic: Perfect Numbers

An inventory-batching system labels a batch size 'balanced' if the sum of all its proper divisors (excluding itself) equals the number itself. Check whether a given batch size is balanced.

INPUT

28

OUTPUT

true ($1 + 2 + 4 + 7 + 14 = 28$)

Q4. Does This ID Loop to Happiness?

Medium

Topic: Happy Numbers

A gamified rewards system marks a user ID 'happy' if repeatedly replacing it with the sum of the squares of its digits eventually reaches 1 (rather than looping forever). Determine whether a given ID is happy.

INPUT

19

OUTPUT

true

Q5. Flip the Transaction Code

Easy

Topic: Reverse Integer

A legacy billing system stores transaction codes as integers and needs them digit-reversed for a checksum step. Reverse the digits of a given integer, keeping the sign, and flag it as invalid if the result overflows a 32-bit signed integer.

INPUT

-1203

OUTPUT

-3021

Q6. Split the Bill Evenly

Medium

Topic: GCD

A group-payment feature needs the greatest common divisor of two spending amounts so it can propose the smallest 'unit' both amounts can be evenly split into.

INPUT

48, 18

OUTPUT

6

Q7. Sync the Delivery Cycles

Medium

Topic: LCM

Two delivery trucks restock on independent cycles measured in days. Find the least common multiple of their cycle lengths to know how many days until both trucks restock on the same day again.

INPUT

4, 6

OUTPUT

12

Q8. Flag Every Prime Under the Limit

Medium

Topic: Sieve of Eratosthenes

A cryptography teaching tool needs every prime number up to a given limit N, generated efficiently rather than by testing each number individually one at a time.

INPUT

`N = 20`

OUTPUT

`[2, 3, 5, 7, 11, 13, 17, 19]`

Q9. Compute Interest, Fast

Medium

Topic: Fast Exponentiation

A finance engine needs to compute base raised to a large exponent, modulo a given number, without the naive loop being too slow when the exponent is in the billions.

INPUT

`base = 2, exponent = 10, mod = 1000`

OUTPUT

`24`

Q10. Build the Seating Triangle

Easy

Topic: Pascal Triangle

An event-seating tool generates the first N rows of Pascal's Triangle to compute combination counts for table arrangements. Print the first N rows.

INPUT

`N = 5`

OUTPUT

`[1]
[1, 1]
[1, 2, 1]
[1, 3, 3, 1]
[1, 4, 6, 4, 1]`

Q11. The Efficient Growth Sequence

Medium

Topic: Fibonacci Optimization

A population-modeling script needs the Nth Fibonacci number, but N can be very large, so a naive recursive solution is too slow. Compute it efficiently.

INPUT

N = 40

OUTPUT

102334155

Q12. Translate the Invoice Number

Medium

Topic: Integer to Roman

A boutique invoicing system stylizes invoice numbers as Roman numerals for print copies. Convert a given integer into its Roman numeral representation.

INPUT

1994

OUTPUT

MCMXCIV

Q13. Decode the Antique Ledger

Medium

Topic: Roman to Integer

A digitization project is converting a scanned antique ledger where quantities are written as Roman numerals back into standard integers.

INPUT

MCMXCIV

OUTPUT

1994

Q14. The Unbreakable Encryption Check

Hard

Topic: Number Theory

A key-generation tool needs to check whether two large numbers are coprime (their greatest common divisor is 1), a required property before they can be used together as an encryption key pair.

INPUT

35, 64

OUTPUT

true

Q15. Count the Trailing Vault Zeros

Hard

Topic: Mathematical Logic

A cryptographic vault uses $N!$ (factorial) as part of its key space and needs to know how many trailing zeros $N!$ has, without ever actually computing the full factorial (N can be enormous).

INPUT $N = 100$ **OUTPUT**

24